UNIVERSITY OF SINDH

JAMSHORO SINDH, PAKISTAN

B. Ed. 04 Year

Course No. GQUR-300 Credit Hours **3**

Title of Course **Quantitative Reasoning-1(Maths)** Semester : 1st

Course Description

This course develops prospective teachers' quantitative reasoning skills required for academic study, professional practice, and everyday problem solving. It emphasizes logical thinking, numerical sense, data interpretation, and the application of mathematical concepts in real-life contexts. The course strengthens foundational mathematical understanding and prepares future teachers to model quantitative reasoning skills for their students.

Learning Outcomes:

The course aims to enable prospective teachers to:

- Demonstrate numerical and algebraic reasoning
- Solve real-world quantitative problems
- Interpret tables, graphs, and statistical information
- Apply proportional and logical reasoning
- Communicate mathematical ideas clearly

Course Outline

Unit 1: Foundations of Quantitative Reasoning

- Nature and importance of quantitative reasoning
- Numeracy and number sense
- Estimation and approximation
- Logical reasoning and problem-solving strategies

Unit 2: Numbers and Operations

- Whole numbers, integers, fractions, and decimals
- Ratios, proportions, and percentages
- Powers and roots
- Applications in daily life

Unit 3: Algebraic Reasoning

- Variables and algebraic expressions
- Linear equations and inequalities
- Patterns and sequences
- Use of algebra in problem solving

Unit 4: Measurement and Geometry

- Units of measurement
- Perimeter, area, and volume

- Basic geometric shapes and properties
- Applications of measurement in real contexts

Unit 5: Data Analysis and Statistics

- Collection and organization of data
- Tables, graphs, and charts
- Measures of central tendency
- Interpretation of statistical information

Unit 6: Quantitative Reasoning in Everyday Life

- Financial literacy (profit, loss, interest, budgeting)
- Quantitative reasoning in media and advertisements
- Interpreting quantitative information in education and society
- Decision-making using quantitative data

Teaching–Learning Strategies:

- Interactive lectures
- Problem-solving sessions
- Group activities
- Real-life case studies
- Guided practice and discussions

Assessment & Evaluation:

- As per University Policy

Recommended Books / References:

1. Steen, L. A. Mathematics and Democracy: The Case for Quantitative Literacy
2. Grawe, N. Quantitative Literacy: Reasoning about Data
3. National Curriculum for Mathematics (Pakistan)
4. HEC Pakistan. National Professional Standards for Teachers

TEACHING/INSTRUCTIONAL STRATEGIES

This course will be covered with the help of following teaching strategies

1. Lecture
2. Discussions
3. Question-Answer
4. Brain storming

PRACTICAL WORK

The student-teachers will also complete their course with the help of following practical work.

1. Group work
2. Group assignments
3. Individual assignment
4. Presentations

ASSESSMENT CRITERIA

The students will be assessed through,

1. Class test 2. Mid Term test
3. Evaluation of assignments 4.Class attendance
5. Class participation/Performance e.g. initiative, responding, curiosity, inquiry